On the Universal Structure of Maximal Planar Graphs

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Abstract

This paper proves that for any maximal planar graph G of order n, if G contains a maximal outerplanar graph of order n as a subgraph, then there exists a homomorphism from G to K_4. Furthermore, this class of four-colorable maximal graphs constitutes the fundamental structure of all maximal graphs, thereby allowing any maximal planar graph to be recursively decomposed into this particular type of maximal planar graph.

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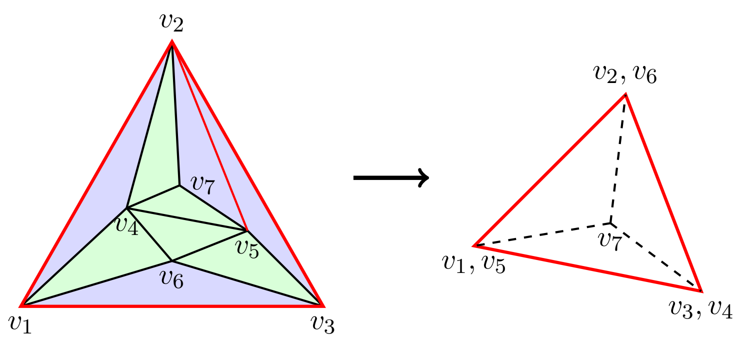
First, we illustrate the overall structure of this work with a concrete example. We then prove that it holds for all natural numbers.

As shown in the figures below.



G_1

G_1 is divided into two parts, marked by blue and green shading. These two parts each having the same number of vertices 7, edges 11, and faces 6.

Consider the green subgraph in G_1. Clearly, it can be **folded** into K_4. And the number of ways to fold it is 2^4. We will prove that one of these ways of folding defines a homomorphism from G_1 to K_4.

This process involves only two difficulties. First, whether the red edges in G_1 may collapse to a single vertex. Second, assuming the first condition is satisfied, whether there exists a vertex that cannot be folded into K_4.

When folding a K_3 along one of its edges, the third vertex always has two possible choices. **Regarding the existence of the red edges in G_1 as constraints. We will prove that if the green subgraph in G_1 cannot be folded into K_4 then there is a K_5 exits in G_1 which contradicts the fact that G_1 is a planar graph.** This resolves the two difficulties mentioned above.

Furthermore we shall prove that every maximal planar graph admits a decomposition into smaller maximal planar graphs via the structure described above, analogous to the decomposition of solvable groups.

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We now generalize the conclusion of the first part to all natural numbers. We begin by recursively defining the maximal planar graphs considered in the first part.

Definition 1. *as follows*

1. *S_0 and R_0 are the same K_3.*
2. *S_1 is obtained from S_0 by adding a vertex inside R_0 and connecting it to the two endpoints of an edge of R_0. R_1 is the quadrilateral in S_1.*
3. *S_n is obtained from S_{n-1} by adding a vertex inside R_{n-1} and connecting it to the two endpoints of an edge of R_{n-1}. R_n is the $(n+3)$-gon in S_n.*

Definition 2. *as follows*

By adding n edges to R_n, we obtain a maximal outerplanar graph, denoted by $\overline{R_n}$. Consequently, S_n becomes a maximal planar graph, denoted by $\overline{S_n}$. In particular, we have $\overline{S_1} = K_4$

From the definitions, S_n and $\overline{R_n}$ each have $n+3$ vertices, $2n+3$ edges, and $n+2$ faces. We have

Theorem 1. *For any $\overline{S_n}$, there exists a homomorphism from $\overline{S_n}$ to K_4.*

Proof. Assume that for all 2^n ways of folding $\overline{R_n}$, there always exists a vertex that cannot be folded into K_4 due to the restriction of $\overline{S_n}$. We shall prove that $\overline{S_n}$ contains a K_5.

Consider an arbitrary folding $f : \overline{R_n} \to K_4$. Suppose that a vertex v cannot be folded into K_4 under f. For every vertex x in K_4, there exists a set X such that every element of X is a preimage of x and is adjacent to v. Since each vertex has two choices during the folding process, some elements in X under f can still be moved to other vertices of K_4 without creating conflicts. We denote by X' the set consisting of those vertices that **cannot** be moved under f. By assumption, X' is nonempty.

The elements of X' have the following property: **every element of X' is adjacent to some element in each of the other three sets X'.** In this way, all four sets X' form a cyclic contraction mapping in terms of the adjacency of their elements.

$$X_1' \to X_2'' \to X_3'' \to X_4'' \to X_1'' \to X_2''' \to X_3''' \to X_4''' \to X_1''' \to \cdots$$

Eventually, these four sets converge to at least one element each, namely $a \in X_1$, $b \in X_2$, $c \in X_3$, and $d \in X_4$. Moreover, a, b, c and d are pairwise adjacent. Hence, *abcdv* forms a K_5.

Thus, among the 2^n ways of folding $\overline{R_n}$, at least one admits a homomorphism from $\overline{S_n}$ to K_4; otherwise, $\overline{S_n}$ contains a K_5.

$\square$

Furthermore, we have

Theorem 2. *For any maximal planar graph G whose order is bigger than 3, there exists a subgraph $\overline{S_n}$ such that $n > 0$ and $\overline{S_n}$ contains the three outermost vertices of G.*

Proof. We denote by $V'(G)$ the set consisting of all vertices of G **except its three outermost vertices**. Since G is a maximal planar graph, for any one of its edges, there exist several vertices in $V'(G)$ such that each of these vertices is adjacent to the two endpoints of that edge, thereby forming a sequence of $K_3's$ that are nested by inclusion.

Keep the largest one of this sequence of $K_3's$ and delete all vertices and edges in its interior. If this vertex is connected to all three outermost vertices, then the resulting subgraph is $\overline{S_1}$.

Apply the above procedure repeatedly until no small K_3 inside the outermost K_3 contains any interior vertex. The graph obtained in this way is the $\overline{S_n}$ we are looking for. Moreover, it is the smallest one.

$\square$

Combining Theorem 1 and Theorem 2, it follows that

The Four Color Theorem holds for all planar graphs.

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We now state the direct condition underlying the above proof. This condition is not that G has no K_5-minor, but that G is planar.

Definition 3. *as follows*

A graph G is said to admit a crossing-free embedding into $\mathbb{R}^{m-2}$ if there exists a mapping of G into $\mathbb{R}^{m-2}$ such that for any $(m-2)$-polytope realized by a subgraph H, no edge of G crosses it.

Using Definition 3, we can easily define maximal graphs in $\mathbb{R}^{m-2}$.

Definition 4. *as follows*

A graph G is called a maximal graph if adding any edge to G destroys the existence of a crossing-free embedding of G into $\mathbb{R}^{m-2}$.

Next, we recursively define maximal outer graphs in $\mathbb{R}^{m-2}$. To do this, we need to use the concept of "gluing" from topology.

Definition 5. *Let $\{u_i\}$ be a set whose elements are K_{m-1}'s. G is called a maximal outer graph, denoted by $\overline{R_n^m}$ or $\overline{R_n}$. Define as follows*

1. *$\overline{R_0}$ is u_0.*
2. *$\overline{R_1}$ is obtained by gluing a K_{m-2} in u_0 and a K_{m-2} in u_1 together from the outside.*

3. $\overline{R_n}$ *is obtained by gluing a* K_{m-2} *that hasn't been glued yet in* $u_{i,\ i<n}$ *and a* K_{m-2} *in* u_n *together from the outside.*

Definition 6. *Normal Graph*

For any maximal graph G*, if* G *contains a subgraph* $\overline{R_n}$ *with the same number of vertices as* G*, then* G *is called a normal graph denoted by* $\overline{S_n^m}$ *or* $\overline{S_n}$*. In particular, if every* K_{m-1} *in* G *has no vertices in its interior, then* G *is called a minimal normal graph.*

Theorem 3. *If every* K_{m-1} *subgraph of a maximal graph* G *has no interior vertices, then* G *is a minimal normal graph.*

Proof. All K_{m-1}'s in G form a finite sequence $\{u_k\}$. One can find a subsequence $\{u_{k_n}\}$ such that the $\overline{R_n}$ formed by $\{u_{k_n}\}$ contains all vertices of G.

□

Theorem 4. *For any* $\overline{S_n}$*, there is a homomorphism from* $\overline{S_n}$ *to* K_m*.*

Proof. Since the proof only involves the properties of $\overline{S_n}$ and $\overline{R_n}$, it can be easily proved in the same way as in the proof of Theorem 1.

□

Theorem 5. *For any maximal graph* G *whose order is bigger than* $m-1$*, and any subgraph* K_{m-1} *with no interior vertices. There exists a subgraph* $\overline{S_n}$ *such that* $n > 0$ *and it contains the* K_{m-1}*.*

Proof. Applying the same procedure as in the proof of Theorem 2 to G, we obtain a minimal normal subgraph of G.

□

By Theorem 4 and Theorem 5, we may conclude that

Theorem 6. *Every graph that admits a crossing-free embedding into* $\mathbb{R}^{m-2}$ *is* m*-colorable.*

We can now give a stronger sufficient condition for a graph to be m-colorable.

Theorem 7. *G is said to be m-colorable if, G can be transformed into a graph admitting a crossing-free embedding into $\mathbb{R}^{m-2}$ by identifying some equivalent vertices.*

Here "equivalent" is defined as, let v^* denote the set of vertices adjacent to v. Vertices v_1 and v_2 are said to be equivalent if $v_1^* \subseteq v_2^*$ and v_1 is not adjacent to v_2.

For instance, $K_{3,3}$ is not a planar graph. But it can be reduced to K_2 in this manner.

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Theorem 5 can also be expressed in algebraic language.

Let G be a maximal graph admitting a crossing-free embedding into $\mathbb{R}^{m-2}$. $\overline{S_n}$ is the **minimal** normal subgraph of G.

Definition 7. *Module of G, denoted by $G^{(j)}$ is defined as follows*

1. *$G^{(0)}$ is $\overline{S_n}$.*
2. *$G^{(j)}$ is the union of $G^{(j-1)}$ and all minimal $\overline{S_n}$ corresponding to the subgraphs associated with each K_{m-1} in $G^{(j-1)}$.*

We have

Theorem 8. *G is a maximal graph admitting a crossing-free embedding into $\mathbb{R}^{m-2}$ if and only if there exists a composition series as follows.*

$$\overline{S_n} = G^{(0)} \triangleright G^{(1)} \triangleright G^{(2)} \triangleright \cdots \triangleright G^{(j)} = G$$

Proof. The sufficiency has already been established in the above definition; we only prove the necessity.

By Definition 7, each newly added part in the composition series is a union of several $\overline{S_n}$, where the outermost K_{m-1} of each $\overline{S_n}$ is exactly a K_{m-1} in some $\overline{S_n}$ from the previous term. Hence, for any $(m-2)$-polytope in G, no edge of G crosses it.

□

References

[1] R. Diestel, *Graph Theory*, 5th ed., Springer, Berlin, 2017.

[2] D. B. West, *Introduction to Graph Theory*, 2nd ed., Prentice Hall, 2001.